**ELEC70076**
**EE9-FPN2-10**

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART IV: MEng and ACGI

**SUSTAINABLE ELECTRICAL SYSTEMS**

Thursday, 8 May 2025, 10:00 AM

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : E. Spyrou, G. Strbac

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# Answer all questions

1. **Question on wind and solar PV power plants**

a) A 8 MW wind turbine is installed at a site you manage. The wind turbine parameters are given in Table 1a and the density of air can be assumed to be 1.22 kg/$m^3$.

Table 1a. Parameters of the wind turbine

| **Parameter** | **Value** |
|---|---|
| Length of turbine blade | 85 m |
| Cut-in speed | 4 m/s |
| Cut-out speed | 25 m/s |
| Rated speed | 11 m/s |

i) Estimate the total efficiency of the turbine.

[2]

ii) Explain in one or two sentences why the total efficiency of a wind turbine will be different to the Betz limit.

[2]

iii) Draw the power curve of the turbine (i.e., graph with wind speeds on x axis and electric power output on the y axis) including appropriate labels that indicate the operating regions.

[2]

iv) Using dashed lines, draw the power curve for the same turbine retrofitted with electric components that allow its rated capacity to increase to 10 MW.

[2]

v) Assuming that the cumulative distribution function of the wind speed $v_w$ at the site is given by $1 - e^{-\left(\frac{v_w}{8}\right)^{2.2}}$, estimate the number of hours in a year that the wind turbine will generate electricity at the rated power 10 MW.

[2]

b) Assume that the sustainability team at Imperial is considering installing solar PV panels on top of flat roofs at the South Kensington campus (latitude ~ 51°). They have access to irradiance data from two different sources, hereafter called Source 1 and Source 2. The data from sources 1 and 2 are provided in Tables 1b1 and 1b2, respectively.

Table 1b1. Irradiance data from Source 1

| | Source 1 |
|---|---|
| Global Horizontal Irradiance (GHI) on 21/12 solar noon | 0.171 $kW/m^2$ |
| Diffuse Horizontal Irradiance on 21/12 solar noon | 0.113 $kW/m^2$ |
| Annual GHI | 1,203 $kWh/m^2$ |

Table 1b2. Irradiance data from Source 2

| | Source 2 |
|---|---|
| DNI on 21/12 solar noon | 0.089 $kW/m^2$ |
| Annual Normal Irradiance (Direct and Diffuse) | 1,752 $kWh/m^2$ |
| Annual electricity generation by double axis tracking 1 kW PV system | 1,438 $kWh$ |

i) Estimate the DNI at solar noon on 21st December using data from source 1. The declination angle is $-23.5°$ that day. Remark in one sentence if the two sources seem consistent.

[2]

ii) Estimate the capacity factor for a double axis tracking 1 kW PV system.

[2]

iii) Assuming that the efficiency of the PV system is 0.25, estimate the area of the solar PV modules for a double axis tracking 1 kW PV system.

[2]

iv) Estimate an upper bound for the capacity factor of a 1 kW PV single-axis tracking system.

[2]

v) Estimate a lower bound for the capacity factor of a 1 kW PV single-axis tracking system.

[2]

## 2. Question on Balancing, Voltage Control and Security of Supply

a) In a specific time slot, the system demand is 25 GW, wind power generation is expected to produce 12 GW and nuclear generation is scheduled to produce 8.4 GW. There is a need for reserves of 6.5 GW, given uncertainty related to wind production. CCGT units of 550 MW capacity with minimum stable generation of 300 MW, are scheduled to provide reserves of 6.5 GW. Determine the total generation production and the amount of wind generation curtailed. Explain the impact on costs and carbon emissions.

[4]

b) What is the State Estimation module in DMS controllers used for? What are Pseudo measurements?

[1]

c) A distributed generator is connected to a 33 kV circuit of the length of 20 km that is connected to load tap changing transformer, that is set to maintain the voltage at the transformer end at the statutory minimum value (statutory voltage limits in 33 kV network are ±6%). Circuit resistivity is 0.0618 Ohms/km and reactivity is 0.107 Ohms/km. Reactive power of distributed generation goes in the opposite direction to the active power.

i) Sketch the considered system with denoted relevant variables and corresponding directions.

[1]

ii) Determine the maximum active power of the distributed generation with 0.95 (inductive) power factor, that could be connected to the end of the 33 kV circuit, when an on-load tap changing transformer is set

to maintain the voltage at the transformer end at the statutory minimum value.

[3]

d) Contribution of distributed generations to the security of supply depends on the demand profiles and availability of the resources. In this case, three units, connected to busbars A can secure a proportion of the demand supplied from the busbar A as shown in Figure 2.1. Those units have ratings of 5, 2.5 and 2.5 MW and availabilities of 90, 80% and 80%, respectively.

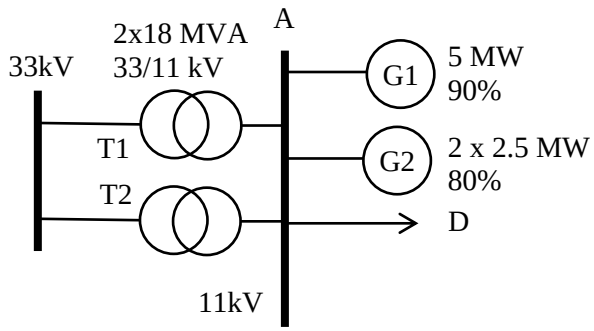



**Figure 2.1 Diagram of the considered part of the network**

Demand profile during a peak season, that lasts 120 days, is represented by the load duration curve as shown in Figure 2.2.

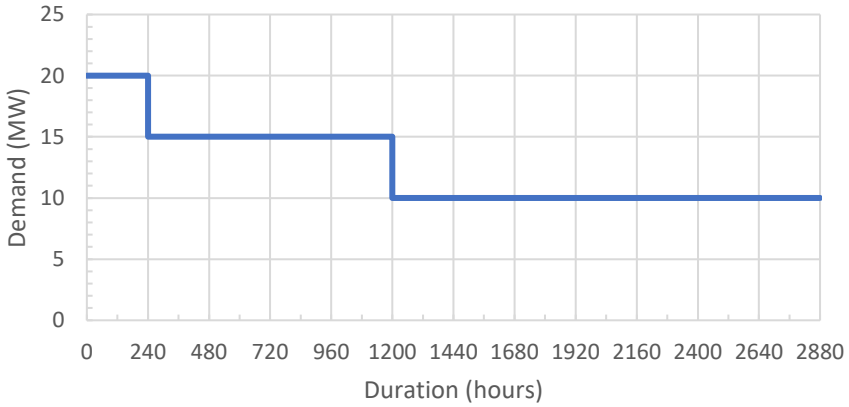



**Figure 2.2 Load duration curve for 120 days**

i) Form the capacity outage probability table for plants providing network support at busbars A by stating capacity in service and the corresponding probability value for each state.

[4]

ii) Calculate the Expected Energy Not Supplied of the system supplied by the generation plants using the load duration curve of Figure 2.2 assuming that 50% of the load is supplied by the distributed generators.

[5]

iii) Calculate the capacity of the equivalent source, i.e. firm capacity, that never fails and results in the same Expected Energy Not Supplied.

[2]

3. **Question on Low Carbon Heat**

a) List the factors that determine building space heat demand. [2]

b) Air source Heat Pump

i) Briefly describe the operation of an air source heat pump (ASHP). [4]

ii) Explain how an ASHP can have an efficiency more than 100%. [1]

iii) What is the difference between an ASHP and a hybrid ASHP? [1]

iv) A householder has decided to purchase an air source heat pump to replace their gas boiler and has a choice of a 5 $kW_{th}$, a 7.5 $kW_{th}$ or a 10 $kW_{th}$ heat pump. Their house has a floor area of 100 $m^2$ and an average thermal transmittance of 3.75 $W_{th}/m^2K$. They want to make sure that the heat pump is sufficient to maintain a building internal temperature of 20 °C when the outside temperature is 0 °C. Calculate the associated space heat demand and which heat pump should therefore be selected. State any assumptions made.

[4]

v) The householder has been advised that the outside temperature can fall to -5 °C but rather than purchase a larger heat pump they decide to increase the building efficiency through insulation improvements. Calculate the percentage improvement in the building's efficiency to avoid purchasing a larger heat pump.

[2]

vi) The average outside temperature is 7 °C and in a typical year the space heating is on for 200 days and 10 hours each day. Calculate the annual space heat demand in a typical year in MWh with the insulation improvements.

[4]

c) The UK Government established the Committee on Climate Change (CCC) in 2008.

i) Briefly summarise the purpose of the CCC. [1]

ii) What amendment did the UK Government make in 2019? [1]

4. **Question on demand-side flexibility and storage**

a) Explain why Electric Vehicle (EV) charging is a potential source of flexibility to support integration of low-carbon sources in future electricity systems. Discuss the implications for energy and peak demand for both smart and non-smart EV charging approaches.

[2]

b) A feeder supplying a residential area in the local LV distribution grid has a baseline (non-EV) demand profile peaking at 245 kW on a winter day (including a few hours of the following night), as shown in Figure 4A. Due to favourable government incentives, 40 EVs have been purchased by residential customers in the area. As specified in Table 4.a, they can be divided into 3 groups according to their driving patterns i.e. the time they arrive home and State of Charge (SoC) at arrival. Each EV has a 25 kWh

battery, while the rating of each EV charger is 5 kW (for both charging and discharging, if allowed). Demand can be assumed constant in each 1-hour period. Ignore any charging or discharging losses. Answer the following questions:

i) If all EV users plug in their vehicles to charge immediately upon arrival, and charge them at maximum capacity until their batteries are full, what is the total peak demand on the feeder and when does it occur? Plot the new total demand diagram.

[2]

ii) By how many hours (at least) do all EVs need to delay their charging after arrival in order to avoid increasing the total peak demand of the feeder above the baseline peak demand of 245 kW? Assume that the same time delay for charging applies to all EVs.

[3]

iii) Assume that in addition to smart charging, EV Group #1 is also willing to provide V2G services (i.e. discharge the batteries) between 17:00 and 20:00. By how much can the total peak demand of the feeder be reduced? For the purpose of this question assume that all EVs across all groups shift their charging after 3:00 am without any impact on the total peak demand of the feeder, i.e., that no EV charging takes place during the time window depicted in Figure 4A. Plot the resulting chart.

[3]

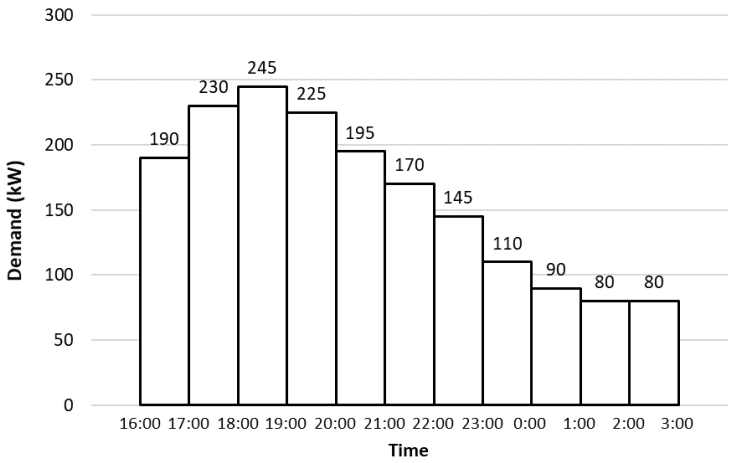



**Figure 4A. Distribution feeder baseline (non-EV) demand**

**Table 4.a EV driving patterns and SoC across 3 vehicle groups**

| Group No. | Number of vehicles | Arrival time | SoC on arrival |
|---|---|---|---|
| 1 | 10 | 17:00 | 80% |
| 2 | 20 | 18:00 | 60% |
| 3 | 10 | 19:00 | 40% |

c) Two UK companies with long duration storage technology are Sunamp and EDF for project StrataStore. Sunamp's patented Plentigrade technology stores energy in high-performance phase change materials and releases it to give hot water, heating or cooling on demand. StrataStore will develop storage of electricity as compressed air, potentially using mothballed EDF owned gas cavities in Cheshire,

i) Please identify for each of the two technologies, the category and sub-category of storage they belong to.

[4]

ii) State two advantages and two disadvantages of the technology used in StrataStore.

[2]

iii) Assuming that the efficiency for the compression stage is 80% and the efficiency for the expansion stage is 80%, what will be the round-trip efficiency of the StrataStore system? How does this efficiency compare to the efficiency of batteries?

[2]

iv) Over the course of a year, do you expect a higher number of cycles for any batteries connected to the grid or the StrataStore system? Please justify your answer.

[2]